OriGround: Orientation-Aware Neuro-Symbolic Zero-Shot 3D Visual Grounding

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Abstract

Zero-shot 3D visual grounding (3DVG) aims to localize a referred object in a 3D scene without task-specific grounding supervision. Existing training-free methods commonly reason over object categories, centers, and bounding boxes, but these representations do not encode the local reference frames needed to interpret viewpoint-dependent expressions such as left of, behind, or facing the door. We propose OriGround, an orientation-aware neuro-symbolic framework that explicitly models object-centric orientations and infers observer viewpoints for zero-shot 3DVG. OriGround parses each query into a viewpoint-aware symbolic program, estimates object-centric orientation bases from multi-view observations, performs anchor- and observer-specific geometric relation scoring, and constructs a reference-frame-aligned visual prompt for final VLM-based disambiguation. On Nr3D, OriGround improves over the strongest training-free baseline by 8.4 points in overall accuracy, with larger gains of 8.9 points on hard queries and 5.6 points on view-dependent queries. On ScanRefer, OriGround further improves overall Acc@0.5 by 17.3 points over VLM-Grounder.

1 Introduction

3D visual grounding (3DVG) aims to localize target objects in 3D scenes from natural-language queries, enabling applications in augmented and virtual reality (Chen et al., 2020), vision-language navigation (Zhu et al., 2025), and robotic perception (Shi et al., 2026; Liu et al., 2026; Achlioptas et al., 2020; Yuan et al., 2024b; Drozdov et al., 2026; Tan et al., 2026). While many referring expressions can be resolved from object categories, attributes, and metric distances, viewpoint-dependent spatial language remains particularly challenging. For example, the phrase "the chair on the left of the sofa" may refer to the sofa's intrinsic left side, whereas "the chair on the left when facing the sofa" depends on the observer's viewpoint. In both cases, the word "left" cannot be interpreted solely in a fixed world coordinate system. Instead, it requires an explicit reference frame defined by either an anchor object, an observer pose, or both.

Recent zero-shot 3DVG methods have made substantial progress by leveraging large language models (LLMs) and vision-language models (VLMs) for language decomposition, symbolic program synthesis, and visual-semantic reasoning. Since pretrained LLMs and VLMs are not directly designed to operate on raw 3D scenes with metric structure, these methods usually convert a scene into object-centric representations, such as semantic categories, instance identifiers, 3D bounding boxes, and center coordinates (Fang et al., 2024; Yuan et al., 2024b; Mi et al., 2025). Such representations are effective for category matching and many view-independent relations. However, they do not encode the local directional bases needed to distinguish front, back, left, and right with respect to a specific object or observer.

To mitigate viewpoint-related ambiguity, recent VLM-based methods further incorporate 2D views or selected viewpoints into the grounding process (Li et al., 2025; Jin et al., 2025). However, rendered views alone do not explicitly specify the reference frame under which a directional relation should be evaluated. The same scene may support different interpretations depending on whether the relation is world-centric, anchor-centric, or observer-centric. Consequently, viewpoint reasoning may still rely on assumptions that the VLM infers from language, rendering choices, or dataset biases. This limitation becomes especially severe for expressions involving *left*, *right*, *front*, *behind*, *facing the door*, or *with your back to the window*.

To address these limitations, we propose **Ori-**

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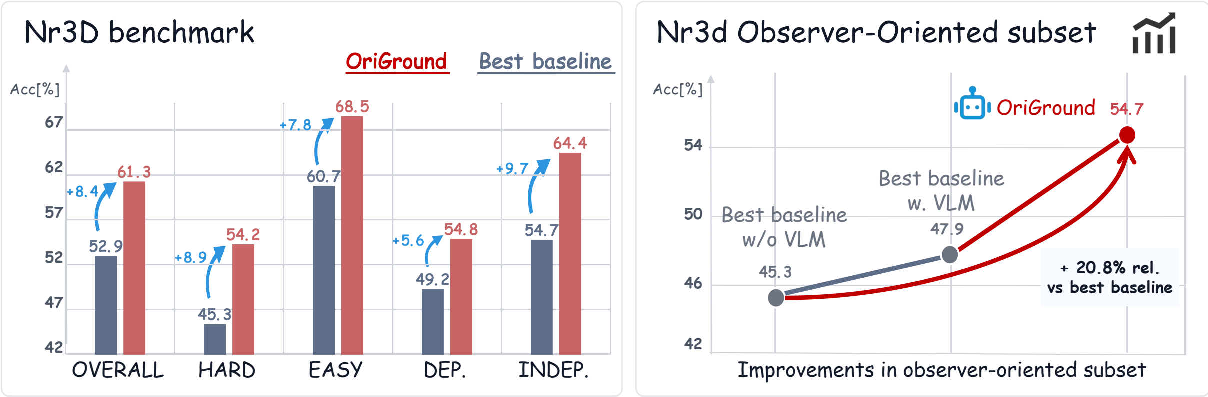

Figure 1: Performance overview of OriGround on Nr3D. Left: comparison with the previous best training-free baseline on standard Nr3D splits. Right: results on the Observer-Oriented subset with explicit observer cues, where OriGround reaches 54.7% accuracy and obtains a 20.8% relative gain.

Ground, an orientation-aware neuro-symbolic framework for zero-shot 3DVG. The key idea is to make the reference frame explicit throughout both symbolic reasoning and visual prompting. OriGround first parses the natural-language instruction into a viewpoint-aware symbolic program that identifies the target category, spatial relations, anchor objects, and observer cues. It then estimates object-centric orientation bases from multi-view object observations and aligns them with the world coordinate system. These orientations define intrinsic up, left, and front directions for each object instance, allowing directional predicates to be evaluated in anchor-specific or observer-specific reference frames. Finally, OriGround constructs a reference-frame-aligned visual prompt that combines multi-view candidate evidence with a top-down map rotated to the inferred observer perspective, enabling the VLM to perform final disambiguation under the same reference frame used by the symbolic executor.

Experiments on Nr3D and ScanRefer demonstrate that OriGround improves over strong training-free baselines. As summarized in Fig. 1, OriGround achieves an 8.4-point overall gain over the previous best training-free baseline on Nr3D, with particularly large improvements on hard, view-dependent, and observer-oriented queries. These results suggest that explicit orientation modeling is not merely an auxiliary visual cue but a useful geometric prior for defining the reference frames under which spatial language should be interpreted.

In summary, the main contributions of this work are as follows:

- We formulate viewpoint-dependent zero-shot 3D visual grounding as a reference-frame ambiguity problem, where bounding-box-only scene representations are insufficient for interpreting directional relations such as left, right, front, and behind.
- We propose an orientation-aware neuro-symbolic reasoning framework that estimates object-centric 3D orientation bases and uses them to score spatial relations in anchor-specific and observer-specific reference frames.
- We introduce reference-frame-aligned visual prompting, which combines multi-view candidate evidence with a dynamically rotated top-down map, and demonstrate consistent improvements over strong training-free baselines on Nr3D and ScanRefer.

2 Related Work

Zero-shot 3D Visual Grounding. Zero-shot 3DVG aims to localize referred objects in 3D scenes without task-specific supervision, reducing reliance on costly 3D grounding annotations (Yuan et al., 2024b; Yang et al., 2024). Existing training-free methods mainly leverage pretrained LLMs or VLMs. LLM-based methods convert 3D scenes into textual or symbolic representations and perform tool-augmented spatial reasoning (Yang et al., 2024; Yuan et al., 2024b,a; Fang et al., 2024), but they lack fine-grained visual evidence and ignore intrinsic object orientations. VLM-based methods introduce rendered views, scene images, or visual prompts for visual-semantic grounding (Li et al., 2025; Xu et al., 2025; Mi et al., 2025; Jin et al., 2025); however, they still rely largely on VLMs to infer 3D viewpoints, which often leads to errors in viewpoint-dependent relations. In contrast, OriGround explicitly extracts object-centric 3D orientations, uses them for deterministic neuro-symbolic reasoning, and aligns visual prompts with the inferred observer perspective.

Neuro-Symbolic Reasoning. Neuro-symbolic reasoning integrates neural perception with the compositionality and interpretability of symbolic reasoning (Andreas et al., 2016; Mao et al., 2019; Yi et al., 2018). Representative methods, such as the Neuro-Symbolic Concept Learner, ground symbolic programs in learned visual concepts, achieving strong data efficiency and compositional generalization (Mao et al., 2019). Recent works further extend this paradigm to 3D scene understanding and grounding, where NS3D performs compositional reasoning over 3D scene-graph representations (Hsu et al., 2023; Feng et al., 2024), and Transcrib3D and LaSP employ LLMs to parse language into executable spatial programs (Fang et al., 2024; Mi et al., 2025). However, these methods mainly rely on positions and bounding-box relations, leaving object-centric orientations unmodeled. OriGround instead injects intrinsic 3D orientations into neuro-symbolic reasoning for reference-frame-aware relation scoring.

3 Methodology

3.1 Problem Formulation and Overview

Problem Formulation. Zero-shot 3D visual grounding aims to localize the target object in a 3D scene from a natural-language query, without task-specific training. Given a scene $\mathcal{S}$ and a query $\mathcal{Q}$, the model selects the referred target $\hat{T}$ from candidate objects $\mathcal{O} = \{O_i\}_{i=1}^{N}$. Each candidate O_i is represented by its 3D point mask, semantic category, bounding box B_i, and center coordinate $\mathbf{c}_i$. For Nr3D, $\hat{T}$ is evaluated as an object identifier; for ScanRefer, it is evaluated by the corresponding 3D bounding box. Since bounding boxes lack intrinsic object directions, they are insufficient for viewpoint-dependent expressions such as *left of*, *behind*, and *facing*. OriGround augments each object with an orientation basis $\mathbf{R}_i = [\mathbf{v}_u^{(i)}, \mathbf{v}_l^{(i)}, \mathbf{v}_f^{(i)}]$, denoting its intrinsic up, left, and front directions. Based on this, OriGround performs viewpoint-aware parsing, orientation-aware symbolic reasoning, and viewpoint-aligned visual prompting to select the target.

Overall Pipeline. To address the ambiguity from missing object-centric directions, we propose OriGround, an orientation-aware neuro-symbolic framework for zero-shot 3D visual grounding. OriGround introduces object-centric orientations to model view-dependent spatial relations and observer cues. As illustrated in Fig. 2, OriGround follows a four-stage inference flow. First, the Language-to-Symbolic Parser (Sec. 3.2) converts the query $\mathcal{Q}$ into a viewpoint-aware symbolic program $\mathcal{P}$. Second, Object-Centric Orientation Estimation (Sec. 3.3) extracts an orientation basis $\mathbf{R}_i$ for each candidate object O_i. Third, Orientation-Aware Symbolic Reasoning (Sec. 3.4) evaluates spatial relations under the corresponding reference frames and produces geometry-based grounding scores. Finally, Holistic Visual Prompting with Viewpoint Alignment (Sec. 3.5) constructs multi-view candidate evidence and a viewpoint-aligned top-down map for the final VLM decision.

3.2 Language-to-Symbolic Parser

Grounding natural-language queries in 3D scenes is challenging due to relational dependencies and implicit references. To bridge this semantic-to-geometric gap, our neuro-symbolic parser uses an LLM to translate the unstructured query $\mathcal{Q}$ into an executable, structured symbolic program $\mathcal{P}$. This intermediate representation explicitly decouples reasoning steps by encapsulating object categories and their spatial constraints.

Specifically, the symbolic program $\mathcal{P}$ contains three core fields: category, relations, and viewpoint. The category field identifies target or anchor objects, the relations field stores spatial predicates and their anchors, and the viewpoint field indicates whether $\mathcal{Q}$ contains a subjective perspective and, if so, its viewpoint anchor.

Due to the unconstrained nature of human language, raw spatial descriptions often form a long-tail distribution. To alleviate this linguistic ambiguity, we prompt the LLM to map the original terms in $\mathcal{Q}$ into a predefined vocabulary of standardized relation labels for $\mathcal{P}$. To disambiguate viewpoint-dependent queries (*e.g.*, "from the foot of the bed"), the parser explicitly extracts the viewpoint component. This mechanism systematically identifies the presence of an egocentric perspective and isolates its corresponding reference anchor. Ultimately, transforming the unstructured input into this viewpoint-aware hierarchical program empowers the downstream modules to compute geometric scores, execute spatial reasoning, and explicitly predict the observer's viewpoint for the VLM.

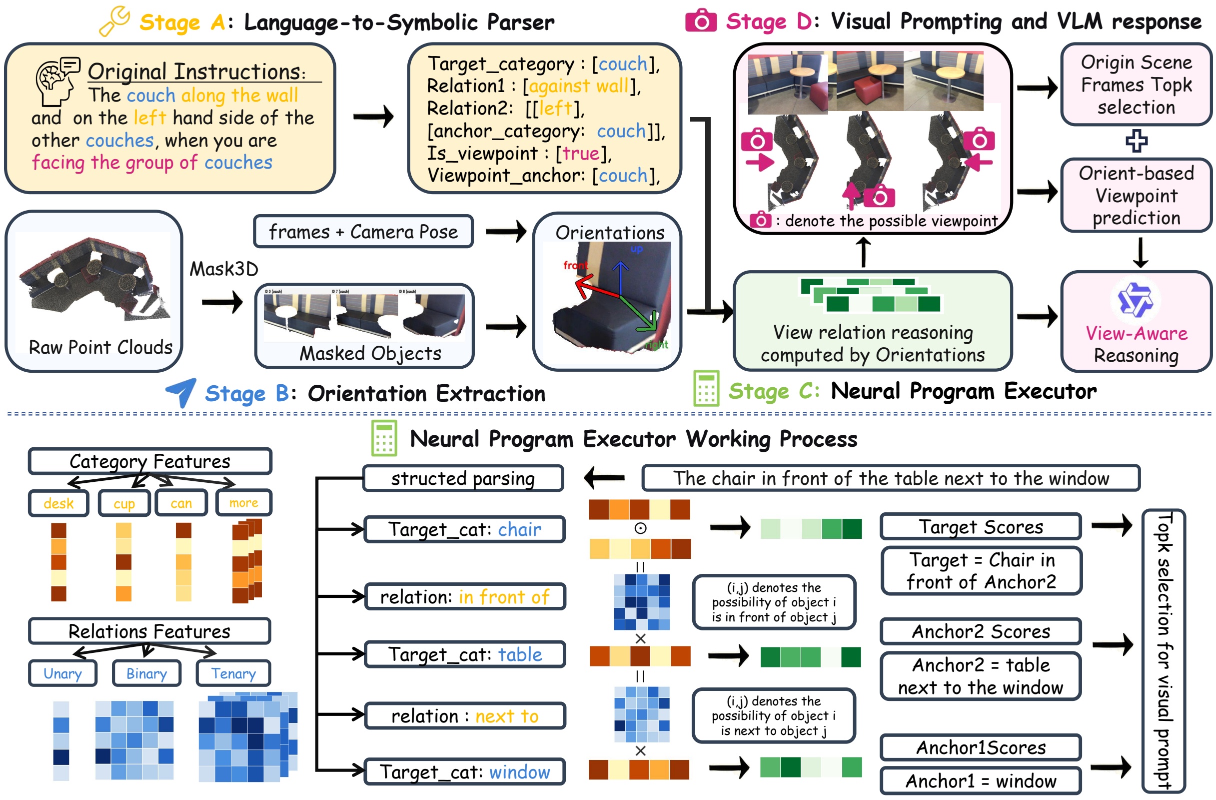

Figure 2: Overview of **OriGround**. Stage A parses the natural-language instruction into a viewpoint-aware symbolic program. Stage B extracts object-centric orientations from masked multi-view observations. Stage C executes the neural-symbolic program to score relations and filter candidates. Stage D builds a viewpoint-aware visual prompt and lets the VLM make the final grounding decision.

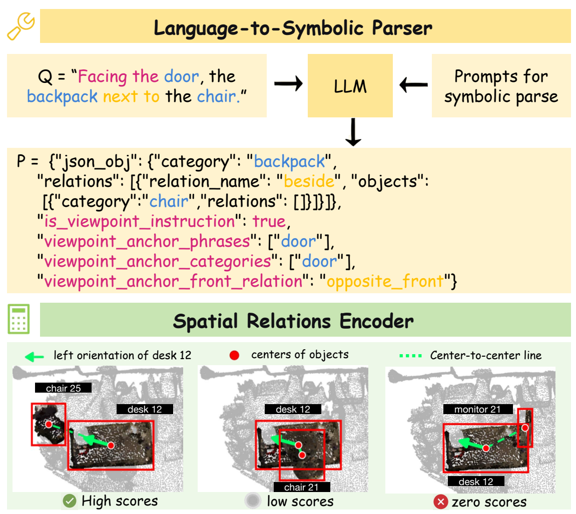

Figure 3: Language-to-symbolic parsing and spatial relation encoding. Top: the parser converts a natural-language query into a structured symbolic program. Bottom: the encoder computes relation scores from object coordinates and intrinsic anchor orientations.

3.3 Object-Centric Orientation Estimation

Orientation Inference and Global Alignment. Orientations, sizes, and positions inherently determine spatial relations. For each object O_i, we isolate its visual appearance in multi-view images using object-level masks projected from the 3D instance mask. The masked image $\hat{I}_i^{(j)}$ is processed by Orient-Anything-V2 (Wang et al., 2025), which predicts a local rotation matrix $\mathbf{R}_{\text{local}} \in \mathrm{SO}(3)$ representing the object's orientation relative to the camera's perspective. When multiple views are available, we use the view with the largest visible object area for orientation inference.

We extract the canonical front vector $\mathbf{n}_f \in \mathbb{R}^3$ from $\mathbf{R}_{\text{local}}$ and transform it into the world coordinate system using camera extrinsic rotation $\mathbf{R}_{\text{ext}}^{(j)}$:

$$\mathbf{v}_{f,\text{raw}}^{(i)} = \mathbf{R}_{\text{ext}}^{(j)} \mathbf{n}_f^{(i)}$$

Considering that indoor objects are typically gravity-aligned, we fix the up vector as $\mathbf{v}_u^{(i)} = [0,0,1]^\top$. To eliminate estimation noise in the pitch axis, we project $\mathbf{v}_{f,\text{raw}}^{(i)}$ onto the horizontal plane to obtain the orthogonal front and left vectors:

$$\tilde{\mathbf{v}}_f^{(i)} = \mathbf{v}_{f,\text{raw}}^{(i)} - \left(\mathbf{v}_{f,\text{raw}}^{(i)\top} \mathbf{v}_u^{(i)}\right) \mathbf{v}_u^{(i)},$$

$$\mathbf{v}_f^{(i)} = \frac{\tilde{\mathbf{v}}_f^{(i)}}{\|\tilde{\mathbf{v}}_f^{(i)}\|_2}, \quad \mathbf{v}_l^{(i)} = \mathbf{v}_u^{(i)} \times \mathbf{v}_f^{(i)}.$$

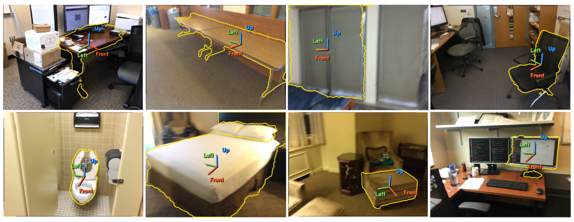

Figure 4: Object-centric orientation extraction examples produced by Orient-Anything-V2.

The resulting object-centric orientations $\mathbf{R}_i = [\mathbf{v}_u^{(i)}, \mathbf{v}_l^{(i)}, \mathbf{v}_f^{(i)}]$ can then be utilized to compute view-dependent geometric relation scores.

Figure 4 visualizes the object-centric front directions estimated by Orient-Anything-V2 for representative scene objects. Each masked object crop is processed by the orientation model, and the predicted front direction is projected back to scene coordinates before symbolic scoring and viewpoint-aligned prompting.

3.4 Orientation-Aware Symbolic Reasoning

Spatial Relations Encoder. For each directional predicate, the executor first determines its reference frame from the symbolic program: world-centric predicates are evaluated in the global frame, anchor-centric predicates in the anchor object's local frame, and observer-centric predicates in the inferred observer frame. The encoder is implemented as executable functions using extracted orientations to compute relational geometric scores. For view-independent relations like *large*, *near*, or *above*, the geometric scores can be directly derived using bounding boxes and center coordinates of the objects. However, these absolute measurements are insufficient for view-dependent relations such as *left* or *behind*. Such relations strictly require the explicit orientation information extracted in our previous module.

Taking the "left" relation as an example (as conceptually illustrated in Fig. 3), let $\mathbf{c}_i, \mathbf{c}_j \in \mathbb{R}^3$ denote the spatial centers of candidate O_i and anchor O_j, respectively. Utilizing the anchor's local left-vector $\mathbf{v}_l^{(j)}$ extracted from its orientation $\mathbf{R}_j$, the normalized direction vector $\mathbf{v}_{ji}$ from the anchor to the candidate, and the final geometric score S_{left} are computed as follows:

$$\mathbf{v}_{ji} = \frac{\mathbf{c}_i - \mathbf{c}_j}{\|\mathbf{c}_i - \mathbf{c}_j\|_2},$$
$$S_{\text{left}}(i,j) = \mathrm{ReLU}(\mathbf{v}_{ji} \cdot \mathbf{v}_l^{(j)}).$$

This formulation evaluates the anchor-centric *left* relation in the local coordinate frame of the anchor object, ensuring that spatial reasoning is consistent with the anchor orientation. For observer-centric directional relations, we replace the anchor-local direction vector with the corresponding direction vector from the inferred observer frame; for instance, $\mathbf{v}_l^{(j)}$ is replaced by the observer-left direction $\mathbf{v}_l^{\text{obs}}$.

Algorithm 1 Recursive Executor

Require: Symbolic node $\mathcal{N}$, category features C, relation features R
Ensure: Grounding score s
$c \leftarrow \mathcal{N}$["category"]
$s \leftarrow C[c]$
for each $rel \in \mathcal{N}$["relations"] **do**
  $r \leftarrow rel$["name"]
  $a \leftarrow rel$["anchor"]
  if a is a nested symbolic node **then**
    $s_a \leftarrow \text{Execute}(a, C, R)$ {Recursive top-down descent}
  else
    $s_a \leftarrow C[a]$ {Base leaf anchor}
  end if
  $\mathbf{M}_r \leftarrow R[r]$
  $s \leftarrow s \odot (\mathbf{M}_r \times s_a)$ {Bottom-up score propagation}
end for
return s

Neural Program Executor. While the parser decomposes queries top-down from the primary target to leaf anchors, geometric scoring propagates strictly bottom-up. Starting from the leaf nodes, each anchor's grounding score s_a is multiplied by its spatial relation matrix $\mathbf{M}_r$, which is computed by the Relation Encoder. This contextual score is then element-wise multiplied (denoted as $\odot$) by the target's initial category score s. By propagating these scores bottom-up, we step-by-step verify all spatial constraints and filter out mismatched candidates and anchors, as detailed in Algorithm 1.

Leveraging the final computed scores, we isolate the top-K candidates along with their high-confidence anchors, paving the way for the subsequent VLM prompt construction and viewpoint prediction.

3.5 Holistic Visual Prompting with Viewpoint Alignment

To leverage the VLM's proficiency in fine-grained feature discrimination and to reduce visual ambiguities through viewpoint prediction, we construct a hybrid prompt comprising instance-level scene images augmented with a viewpoint-aligned global context as illustrated in Fig. 5.

Multi-View Instance Prompting (MIP). After pruning the search space to the top-K candidates, we build a compact 2×2 visual prompt for each can-

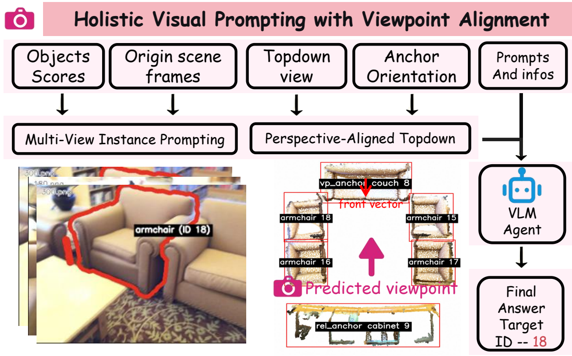

Figure 5: Holistic visual prompting with viewpoint alignment. The VLM receives candidate instance evidence together with a perspective-aligned top-down map to synthesize a final response.

didate. One view is the frame where the candidate occupies the largest visible area, while the other is chosen to maximize co-visibility with the relevant anchor whenever possible. The top row keeps the original frames, and the bottom row shows the corresponding annotated views, giving the VLM both clear candidate appearance and explicit local relational evidence.

Perspective-Aligned Global Context (PGC). To provide a holistic spatial layout, we render a top-down map $\mathcal{M}_{\text{td}}$ using only the shortlisted candidates and the relation-relevant anchors retained by the symbolic executor. This reduces irrelevant clutter while preserving the geometry required by the query. We then rotate the map to the observer perspective θ_{obs}, which is derived from the symbolic program $\mathcal{P}$ and the orientation of the reference anchor. The resulting perspective-aligned map is denoted as $\widetilde{\mathcal{M}}_{\text{td}}$.

Orientation-Hint Rendering. In addition to candidate positions, we render the front directions of relation-relevant anchors on the perspective-aligned top-down map. These orientation hints expose the local reference frames used by the symbolic executor and help the VLM interpret directional relations such as *left*, *right*, and *behind*. We denote the final annotated map as $\widehat{\mathcal{M}}_{\text{td}}$.

Final VLM Decision. The textual query $\mathcal{Q}$, candidate-specific instance prompt $\mathcal{V}_{\text{inst}}^{(i)}$, and annotated top-down map $\widehat{\mathcal{M}}_{\text{td}}$ are jointly fed into the VLM. The final target is selected from the top-K candidates:

$$\hat{T} = \underset{O_i \in \mathcal{O}_{\text{top-}K}}{\arg\max}\, P_{\text{VLM}}\left(O_i \mid \mathcal{Q}, \mathcal{V}_{\text{inst}}^{(i)}, \widehat{\mathcal{M}}_{\text{td}}\right).$$

These inputs allow the VLM to compare shortlisted candidates using both local appearance evidence and global spatial layout.

4 Experiments

4.1 Experimental Settings

Datasets. We evaluate OriGround on two standard 3D visual grounding benchmarks: Nr3D from ReferIt3D (Achlioptas et al., 2020) and ScanRefer (Chen et al., 2020). Nr3D contains 41,503 human-written descriptions and is reported on the *Easy*, *Hard*, *View-dependent*, and *View-independent* subsets, where *Hard* indicates stronger same-category ambiguity and *View-dependent* indicates observer-centric spatial language. Since Nr3D provides ground-truth 3D bounding boxes, evaluation amounts to identifying the correct target object ID. ScanRefer contains 51,583 expressions from 800 ScanNet scenes and reports Acc@0.25 and Acc@0.5 on the *Overall*, *Unique*, and *Multiple* splits. Unlike Nr3D, ScanRefer requires predicting a target 3D bounding box and evaluates grounding accuracy by IoU.

Implementation Details. In all experiments, the language-to-symbolic parser and the visual grounding stage use Qwen-VL-Max from the Qwen-VL family (Bai et al., 2023). Object orientations are estimated with Orient-Anything-V2 (Wang et al., 2025) and projected into the world coordinate system as described in Sec. 3.4. To reduce costs, we randomly select 1,000 validation samples from ScanRefer for testing. We report the performance of the baselines from their original papers. For the analysis in Table 4, we construct the Observer-Oriented subset of Nr3D (Nr3D-VP) using our symbolic parser. It contains 1,859 queries whose symbolic programs include explicit observer cues or viewpoint anchors.

4.2 Comparative Study

Nr3D. Table 1 reports the results on Nr3D. Among training-free methods, OriGround achieves the best performance across all standard splits, improving over LaSP (Mi et al., 2025) by 8.4 points in overall accuracy. The gains are especially clear on the Hard and View-dependent subsets, indicating that explicit orientation modeling and viewpoint-aligned prompting are effective for same-category ambiguity and observer-dependent spatial language. Notably, OriGround also approaches top supervised methods and even exceeds BUTD-DETR in overall accuracy, despite using no task-specific grounding supervision.

Method	Supervised	Overall	Easy	Hard	View Dep.	View Indep.
ViL3DRef (Chen et al., 2022)	✓	64.4	70.2	57.4	62.0	64.5
CoT3DRef (Bakr et al., 2024)	✓	64.4	70.0	59.2	61.9	65.7
3D-VisTA (Zhu et al., 2023)	✓	64.2	72.1	56.7	61.5	65.1
BUTD-DETR (Jain et al., 2022)	✓	54.6	60.7	48.4	46.0	58.0
SAT (Yang et al., 2021)	✓	49.2	56.3	42.4	46.9	50.4
ZSVG3D (Yuan et al., 2024b)	×	40.2	49.1	31.1	37.8	41.6
SeeGround (Li et al., 2025)	×	46.1	54.5	38.3	42.3	48.2
VLM-Grounder (Xu et al., 2025)	×	48.0	55.2	39.5	45.8	49.4
LaSP w/o VLM (Mi et al., 2025)	×	50.7	58.7	43.0	45.6	53.2
LaSP (Mi et al., 2025)	×	52.9	60.7	45.3	49.2	54.7
LaSP + Qwen-VL-Max† (Mi et al., 2025)	×	52.8	61.0	44.9	48.3	55.1
OriGround (ours)	×	**61.3**	**68.5**	**54.2**	**54.8**	**64.4**

Table 1: Comparison on Nr3D. Best training-free results are bolded. The Supervised column denotes whether each method uses task-specific supervised training. † denotes our full-set rerun with Qwen-VL-Max.

Method	Supervised	Overall		Unique		Multiple	
		Acc@0.25	Acc@0.5	Acc@0.25	Acc@0.5	Acc@0.25	Acc@0.5
ScanRefer (Chen et al., 2020)	✓	37.3	24.3	65.0	43.3	30.6	19.8
TGNN (Huang et al., 2021)	✓	34.3	29.7	64.5	53.0	27.0	21.9
InstanceRefer (Yuan et al., 2021)	✓	40.2	32.9	77.5	66.8	31.3	24.8
3DVG-Transformer (Zhao et al., 2021)	✓	47.6	34.7	81.9	60.6	39.3	28.4
BUTD-DETR (Jain et al., 2022)	✓	52.2	39.8	84.2	66.3	46.6	35.1
OpenScene (Peng et al., 2023)	×	13.2	6.5	20.1	13.1	11.1	4.4
LLM-Grounder (Yang et al., 2024)	×	17.1	5.3	–	–	–	–
ZSVG3D (Yuan et al., 2024b)	×	36.4	32.7	63.8	58.4	27.7	24.6
SeeGround (Li et al., 2025)	×	44.1	39.4	75.7	68.9	34.0	30.0
VLM-Grounder* (Xu et al., 2025)	×	51.6	32.8	66.0	29.8	**48.3**	**33.5**
VLM-Grounder‡ (Xu et al., 2025)	×	46.8	28.7	61.8	50.0	38.2	21.6
OriGround (ours)‡	×	**53.0**	**46.0**	**78.1**	**72.1**	38.5	30.9

Table 2: Comparison of 3DVG results on ScanRefer. Best training-free results are bolded. * denotes VLM-Grounder's reported 250-sample result; ‡ denotes the same 1,000-sample evaluation using Qwen-VL-Max.

#	VLM	MIP	PGC	Ori.-Hint	Overall	View Dep.
(a)	×	×	×	×	50.0	44.2
(b)	✓	✓	×	×	58.6	51.2
(c)	✓	✓	✓	×	60.5	53.0
(d)	✓	✓	✓	✓	**61.3**	**54.8**

Table 3: Ablation on Nr3D. Rows progressively add MIP, PGC, and Ori.-Hint to Symbolic Filtering. Overall and View-dependent accuracy (%) are reported.

ScanRefer. Table 2 evaluates OriGround and VLM-Grounder (Xu et al., 2025) on ScanRefer using the same 1,000-sample subset. OriGround obtains the best overall training-free performance, with a particularly large gain on Acc@0.5.

4.3 Ablation Study

Ablation of Architecture Design. Table 3 summarizes the contribution of the main visual-prompting components in Sec. 3.5: Multi-View Instance Prompting (MIP), Perspective-Aligned Global Context (PGC), and the final Orientation-Hint rendered on the top-down map.

- **(a) Symbolic Filtering.** Without the final VLM stage, symbolic filtering alone reaches 50.0% overall accuracy. This shows that symbolic reasoning already removes many inappropriate candidates, but it is not enough for fine-grained disambiguation.
- **(b) Instance-Centric Prompting.** Adding the VLM together with MIP raises accuracy from 50.0% to 58.6%. This contributes the largest gain in the table, showing that candidate visual evidence is crucial after symbolic pruning because it enables the VLM to access fine-grained visual features.
- **(c) Perspective-Aligned Prompting.** Our proposed PGC module improves accuracy from 58.6% to 60.5%. This indicates that a viewpoint-aligned top-down map provides useful global context, complementing the limited scope of local candidate images.
- **(d) Orientation-Hint.** Orientation annotations on the aligned map further improve the grounding accuracy, confirming that explicit orientation cues help resolve directional rela-

Variant	Modules				Observer-Oriented Overall
	VLM	MIP	PGC	Ori.-Hint	
LaSP (symbolic only)	×	×	×	×	45.3
LaSP (+ VLM)	✓	×	×	×	47.9
OriGround (+ MIP)	✓	✓	×	×	50.9
OriGround (full)	✓	✓	✓	✓	**54.7**

Table 4: Perspective-alignment study on the Observer-Oriented subset of Nr3D (Nr3D-VP). The module columns follow the same notation as Table 3.

Method	Model	Overall	Easy	Hard	View Dep.	View Indep.
LaSP	Qwen-VL-Max	50.3	56.2	44.5	45.6	52.4
OriGround	GPT-4o	53.8	62.2	45.7	43.3	58.8
OriGround	Qwen3-VL-Flash	57.0	67.2	46.9	50.0	60.2
OriGround	Qwen-VL-Max	**62.3**	**71.0**	**53.8**	**53.4**	**66.4**

Table 5: VLM model ablation on a shared 1,000-sample Nr3D probe split.

tions and spatial understanding.

Effect of Perspective-Alignment. Table 4 examines grounding performance on Nr3D-VP. Starting from Symbolic Filtering, LaSP (Mi et al., 2025) reaches 45.3%. With its own VLM stage, LaSP (Mi et al., 2025) reaches 47.9%. Under the OriGround architecture, enabling the MIP module raises accuracy to 50.9%, showing that our candidate construction already yields stronger visual evidence on this subset. Adding the PGC module together with Orientation-Hint further lifts accuracy to 54.7%. The gain indicates that aligning the top-down map to the query-consistent observer viewpoint is especially important when the utterance explicitly specifies the observer's direction.

Ablation of Vision-Language Model. Table 5 separates the effects of architecture and backbone on a 1,000-sample probe split randomly sampled from Nr3D to ensure evaluation fairness and representativeness. For a fair architecture comparison, we rerun LaSP (Mi et al., 2025) with Qwen-VL-Max from the Qwen-VL family (Bai et al., 2023); under the same VLM model, replacing LaSP with OriGround raises the overall accuracy from 50.3% to 62.3%, showing that our proposed orientation-aware symbolic pipeline and visual prompting contribute a substantial gain. Keeping the OriGround architecture fixed and then changing the final VLM further affects performance, from 53.8% with GPT-4o (OpenAI, 2024) to 57.0% with Qwen3-VL-Flash from the Qwen3-VL family (Bai et al., 2025), with the best result obtained by Qwen-VL-Max at 62.3% overall. Together, these results indicate that OriGround performs best when a stronger VLM is

Setting	Overall	Easy	Hard	View Dep.	View Indep.
Original	61.70	69.21	54.65	55.72	64.67
Yaw noise	60.60	68.80	52.91	51.81	64.97
Yaw noise w/o VLM	48.20	56.61	40.31	39.16	52.69
180° yaw flip	57.30	63.02	51.94	41.27	65.27
180° yaw flip w/o VLM	44.40	51.24	37.98	27.41	52.84

Table 6: Orientation robustness on a fixed 1,000-sample Nr3D subset. All values are accuracy (%).

paired with the proposed architecture, allowing the structured visual prompt to be exploited more fully.

4.4 Robustness to Orientation Perturbations

We test a fixed 1,000-sample Nr3D subset by perturbing only each object's horizontal front vector; all other inputs and settings remain unchanged. Gaussian noise independently rotates each vector by $\Delta\theta \sim \mathcal{N}(0, (30^\circ)^2)$, clipped to $[-30^\circ, 30^\circ]$. The 180° yaw flip reverses every front vector about the gravity-aligned z-axis. "w/o VLM" reports the symbolic executor's top-1 prediction.

Gaussian noise reduces Overall accuracy by only 1.10 points, but affects View Dep. more strongly (3.91 points). The 180° flip causes a larger View Dep. drop (14.45 points), while View Indep. remains stable, confirming the importance of orientation for viewpoint-dependent queries. The final VLM improves Overall accuracy over symbolic top-1 by 12.40 and 12.90 points under noise and flipping, respectively, by using the retained candidates and visual evidence.

4.5 Qualitative Results

Figure 6 provides qualitative evidence for the gains reported in Table 4. Blue text denotes object categories, yellow text denotes relation terms, and magenta text marks observer-oriented expressions such as *facing the door* or *with your back to the door*. These viewpoint cues define the observer frame under which directional relations should be interpreted.

From the perspective of module design, these examples show why MIP alone is often insufficient on Nr3D-VP: after symbolic filtering, multiple candidates can still remain locally plausible. The PGC module addresses this ambiguity by rotating the top-down map into the query-consistent observer frame, while Ori.-Hint further renders object front directions on the aligned map. Together, these two components let the VLM interpret directional relations under the correct perspective, which qualitatively explains the improvement from the MIP-only system to the full OriGround system.

Figure 6: Qualitative examples from the Observer-Oriented subset of Nr3D (Nr3D-VP). Blue denotes object categories, yellow denotes relation terms, and magenta text highlights observer-oriented expressions.

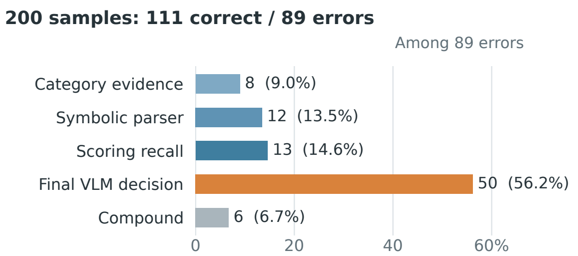

Figure 7: Failure attribution on 200 View-dependent Nr3D samples. Percentages are computed over the 89 errors.

Method	Time/s	Token
ZSVG3D (Yuan et al., 2024b)	2.4	2.5k
VLM-Grounder (Xu et al., 2025)	50.3	8k
Transcrib3D (Fang et al., 2024)	27.0	12k
CSVG (Yuan et al., 2024a)	4.0	4.0k
SeeGround (Li et al., 2025)	9.0	2.6k
LaSP (Mi et al., 2025) (w/o VLM)	2.1	1.2k
LaSP (Mi et al., 2025)	7.7	3.1k
OriGround	10.2	7.1k

Table 7: Per-sample runtime and token usage comparison.

4.6 Stage-Wise Error Analysis

We analyze 200 View-dependent Nr3D samples: 111 are correct and 89 are errors. Each error is assigned to one mutually exclusive category along the pipeline. Category-evidence failures cover missing candidates or cases where the target's category score falls in the bottom quartile. Parser errors are manually verified incorrect symbolic programs; scoring-recall failures have a correct program but exclude the target from the symbolic top-5; and VLM failures retain the target in the top-5 but produce an incorrect or missing final answer. Compound errors involve multiple stages.

Final VLM decisions account for the largest share of errors (56.2%), while category evidence, parsing, and symbolic recall together account for 37.1%. Thus, final disambiguation is the main bottleneck, but upstream errors remain substantial.

4.7 Efficiency Analysis

Table 7 compares the per-sample runtime and token usage of OriGround with prior training-free methods. OriGround uses 7.1k tokens per sample and takes 10.2 seconds end to end, including 4.2 seconds for symbolic processing and 6.0 seconds for visual reasoning.

5 Conclusion

We presented OriGround, an orientation-aware neuro-symbolic framework for zero-shot 3D visual grounding. By parsing language into a viewpoint-aware symbolic program, extracting object-centric orientations, executing orientation-aware geometric reasoning, and aligning the final visual prompt with the inferred observer frame, OriGround resolves view-dependent spatial expressions more reliably than prior training-free approaches. Experiments on Nr3D and ScanRefer show consistent gains over strong training-free baselines, with the largest improvements appearing on observer-oriented queries. These results suggest that explicit orientation modeling is a practical step toward more robust zero-shot grounding in embodied 3D environments. More broadly, this integrated design supports reliable embodied spatial reasoning under ambiguous viewpoints.

6 Limitations

Inference Cost. OriGround combines symbolic parsing, orientation estimation, geometric execution, and a final VLM decision stage. This richer pipeline is still less lightweight than simpler training-free alternatives, and the added symbolic processing plus viewpoint-aligned prompting increase both latency and token usage. More details are provided in Sec. 4.7.

Dependence on Upstream Models. OriGround benefits from reliable candidate generation, language parsing, orientation estimation, and final VLM verification. When these upstream components are noisy, errors can propagate through the pipeline and make the final ranking less stable, especially for fine-grained directional distinctions.

Viewpoint Abstraction. Our perspective-aligned top-down map is most natural for horizontal observer-centric relations. It is less expressive for weakly specified viewpoints, vertically structured cases, or spatial cues that require richer 3D context than a floor-plane abstraction can capture.

Gravity Alignment. We assume that each object's intrinsic up direction is aligned with gravity. This simplification may fail for tilted, side-standing, or otherwise non-canonical objects.

Symbolic Expressiveness. Like LaSP (Mi et al., 2025), OriGround maps open-ended language into a predefined symbolic vocabulary. This keeps reasoning controllable and efficient, but it can be less flexible for rare or highly compositional expressions that do not map cleanly to the supported predicates.

7 Ethical Considerations

This work does not involve new data collection, human-subject experiments, or additional manual annotation. We evaluate only on existing public benchmarks and use third-party pretrained components. OriGround depends on object proposals, language parsing, orientation estimation, and VLM-based verification; these components may inherit biases or failure modes from their underlying data and models. Consequently, upstream errors may propagate to grounding predictions.

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This appendix provides supplementary details for OriGround: Section A presents qualitative reasoning cases, Section B details the experimental setup, and Section D lists the main prompt templates.

A Representative Reasoning Cases

We present four case studies that expose the raw symbolic output, executor scores, visual prompts, and final VLM reasoning evidence.

A.1 Observer-Oriented Example 1

Original Instruction

```
{
  "target_id": 3,
  "sentence": "If you face the
    whiteboard, the chair is the one on
    the left closer to the board."
}
```

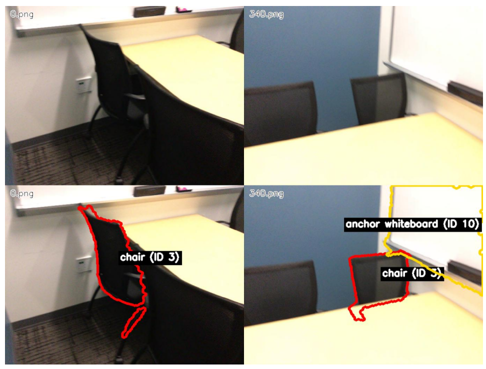
Symbolic Parser Output

```
{
  "json_obj": {
    "category": "chair",
    "relations": [
      {
        "relation_name": "left",
        "objects": [
          {
            "category": "whiteboard",
            "relations": []
          }
        ]
      },
      {
        "relation_name": "closer",
        "objects": [
          {
            "category": "whiteboard",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["
    whiteboard"],
  "viewpoint_anchor_categories": ["
    whiteboard"],
  "viewpoint_anchor_front_relation": "
    opposite_front"
}
```

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	3	1	11	2	4
Cat.	chair	chair	chair	chair	table
Score	1.000	0.870	0.540	0.486	2.23e-6

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.

Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description
    specifies facing the whiteboard and
    identifying the chair on the left
    closer to it. The top-down map shows
     the observer viewpoint (green arrow)
     facing the whiteboard (anchor 10).
    Chair ID 3 is positioned to the left
     of the whiteboard from this
    viewpoint and is closer to it than
    other chairs. The candidate image
    for ID 3 confirms its proximity to
    the whiteboard and its position on
    the left side when facing the board.
     Other chairs (ID 1, 2, 11) are
    either farther or not on the left
    side relative to the whiteboard.",
  "object id": 3
}
```

A.2 Observer-Oriented Example 2

Original Instruction

```
{
  "target_id": 20,
  "sentence": "The lamp to the right if
    you are lying on the bed. The lamp
    closest to the doll house."
}
```

Symbolic Parser Output

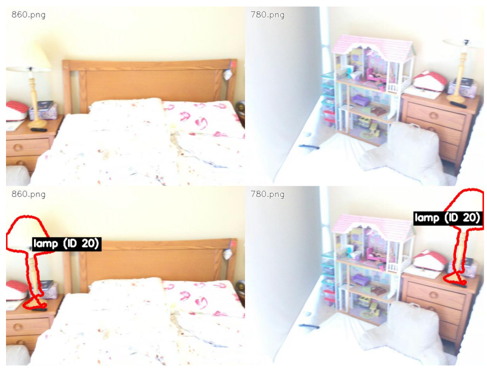
```
{
  "json_obj": {
    "category": "lamp",
    "relations": [
      {
        "relation_name": "right",
        "objects": [
          {
            "category": "bed",
            "relations": []
          }
        ]
      },
      {
        "relation_name": "closest",
        "objects": [
          {
            "category": "doll house",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["bed"],
  "viewpoint_anchor_categories": ["bed"],

  "viewpoint_anchor_front_relation": "
    same_front"
}
```

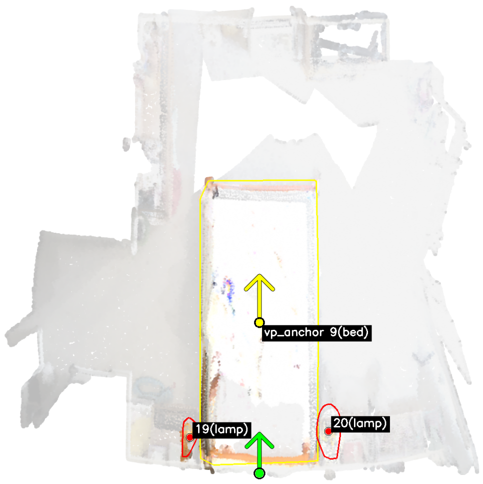
Executor Top-5 Selection

Rank	1	2	3	4	5
ID	19	20	1	0	2
Cat.	lamp	lamp	book	books	trash can
Score	1.000	0.480	5.37e-6	1.23e-6	1.86e-7

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description
    specifies the lamp to the right when
     lying on the bed, and the one
    closest to the doll house. From the
    top-down map, the observer viewpoint
     (green arrow) is facing the bed,
    with the bed's front direction
    indicated by the yellow arrow. Lamp
    ID 20 is positioned to the right of
    the bed from this viewpoint and is
    adjacent to the doll house, as seen
    in the candidate image for ID 20.
    Lamp ID 19 is on the left side of
    the bed. The doll house is clearly
    visible next to lamp ID 20 in the
    annotated views, confirming
    proximity. Thus, lamp ID 20
    satisfies both spatial conditions.",
  "object id": 20
}
```

A.3 Observer-Oriented Example 3

Original Instruction

```
{
  "target_id": 12,
  "sentence": "When standing at the foot
     of the bed, the pillow in the
    middle on the left hand side"
}
```

Symbolic Parser Output

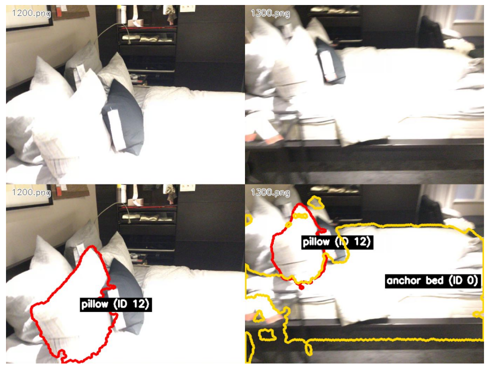
```
{
  "json_obj": {
    "category": "pillow",
    "relations": [
      {
        "relation_name": "middle",
        "objects": [
          {
            "category": "bed",
            "relations": []
          }
        ]
      },
      {
        "relation_name": "left",
        "objects": [
          {
            "category": "bed",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["the foot
     of the bed"],
  "viewpoint_anchor_categories": ["bed"],

  "viewpoint_anchor_front_relation": "
    opposite_front"
}
```

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	12	15	11	13	14
Cat.	pillow	pillow	pillow	pillow	pillow
Score	1.000	0.922	0.788	0.588	0.588

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.

Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description
    specifies standing at the foot of
    the bed and identifying the pillow
    in the middle on the left-hand side.
     The top-down map shows the observer
     viewpoint (green arrow) facing the
    head of the bed, with the foot of
    the bed at the bottom of the map.
    From this perspective, the left side
     of the bed corresponds to the left
    side of the map. Among the pillows,
    ID 12 is positioned centrally along
    the left side of the bed when viewed
     from the foot. The candidate image
    for ID 12 shows it located in the
    middle-left cluster of pillows,
    consistent with the description.
    Other candidates like ID 15 are more
     toward the far left, and IDs 13 and
     14 are on the right side, making ID
     12 the best match.",
  "object id": 12
}
```

A.4 Nested Spatial Relation Example

Original Instruction

```
{
  "target_id": 0,
  "sentence": "black backpack to right
    of the dresser with tv on it nearest
     the opening to the bathroom and the
     end of the bed,"
}
```

Symbolic Parser Output

```
{
  "json_obj": {
    "category": "backpack",
    "relations": [
      {
        "relation_name": "right",

        "objects": [
          {
            "category": "dresser",
            "relations": [
              {
                "relation_name": "above",

                "objects": [
                  {
                    "category": "tv",
                    "relations": []
                  }
                ],
                "negative": true
              },
              {
                "relation_name": "close",

                "objects": [
                  {
                    "category": "opening
      ",

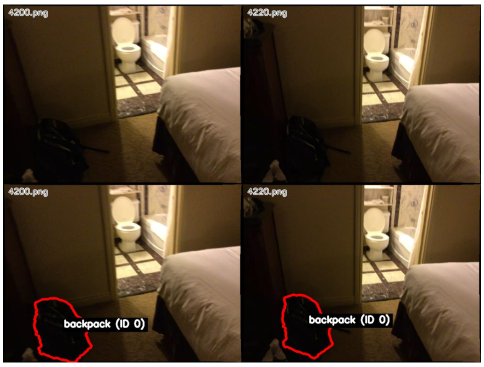
                    "relations": []
                  }
                ]
              },
              {
                "relation_name": "close",

                "objects": [
                  {
                    "category": "bed",
                    "relations": []
                  }
                ]
              }
            ]
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": false,
  "viewpoint_anchor_phrases": [],
  "viewpoint_anchor_categories": [],
  "viewpoint_anchor_front_relation": "
    unknown"
}
```

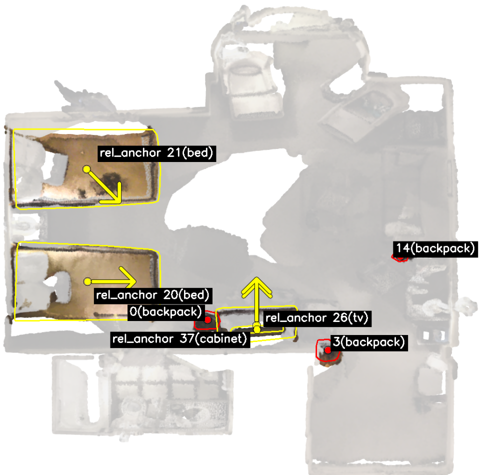
Executor Top-5 Selection

Rank	1	2	3	4	5
ID	0	3	14	1	76
Cat.	backpack	backpack	backpack	bag	toilet paper
Score	1.000	0.951	0.937	7.27e-4	8.86e-7

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. The original query does not contain an explicit observer viewpoint, so the top-down map provides only global layout information.

VLM Reasoning

```
{
  "reasoning": "The target description
    specifies a black backpack to the
    right of a dresser with a TV on it,
    near the bathroom opening and the
    end of the bed. In the top-down map,
     ID 0 is positioned near the bed and
     bathroom entrance, adjacent to the
    cabinet (ID 37) which has a TV (ID
    26) on it. The annotated images for
    ID 0 show a black backpack located
    to the right of the dresser with the
     TV, consistent with the description.
     The other candidates (IDs 3, 14, 1,
     76) are either not near the
    specified anchor objects or do not
    match the spatial relationship
    described. ID 76 is toilet paper,
    not a backpack. Thus, ID 0 best
    matches the target description.",
  "object id": 0
}
```

B Experimental Details

Basic evaluation settings. We keep the top-5 candidates after symbolic execution and set the filtering threshold to 0.0. In the VLM stage, the decoding hyper-parameters are temperature $= 0.1$ and top-$p = 0.2$. For the evaluation on ReferIt3D, we use the object classification results from ZSVG3D (Yuan et al., 2024b).

Observer-Oriented subset validation. We construct Nr3D-VP through rule-based cue detection and LLM screening, followed by manual verification of all 1,859 selected queries. The verification confirms that each query contains explicit observer cues or viewpoint anchors; this subset is used only for analysis and not for training.

The following examples show the VLM recovering targets retained in the symbolic top-5.

B.1 Robustness Recovery Example 1

Original Instruction

```
{
  "target_id": 66,
  "sentence": "Facing the shower stall,
    it’s the bottle on the left."
}
```

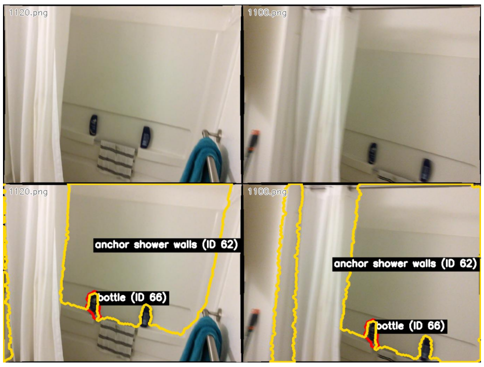
Symbolic Parser Output

```
{
  "json_obj": {
    "category": "bottle",
    "relations": [
      {
        "relation_name": "left",
        "objects": [
          {
            "category": "shower stall",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["shower
    stall"],
  "viewpoint_anchor_categories": ["
    shower stall"],
  "viewpoint_anchor_front_relation": "
    opposite_front"
}
```

Executor Top-5 Selection

Rank	1	2	3	4	5
ID	67	35	22	31	66
Cat.	bottle	jacket	ceiling fan	shoes	bottle
Score	1.000	3.20e-6	1.30e-6	9.76e-7	4.15e-7

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.

Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description
    specifies 'facing the shower stall,
    it's the bottle on the left.' The
    top-down map shows two bottles (ID
    66 and ID 67) near the shower walls
    (anchor ID 62). The green arrow
    indicates the observer is facing the
     shower stall. In the candidate
    images, ID 66 is positioned to the
    left of the shower stall when facing
     it, while ID 67 is on the right.
    The annotated views confirm that ID
    66 is the left bottle relative to
    the shower stall from the observer's
     perspective. Therefore, ID 66
    matches the description.",
  "object id": 66
}
```

B.2 Robustness Recovery Example 2

Original Instruction

```
{
  "target_id": 18,
  "sentence": "it is the bottle on the
    right-hand side of the sink"
}
```

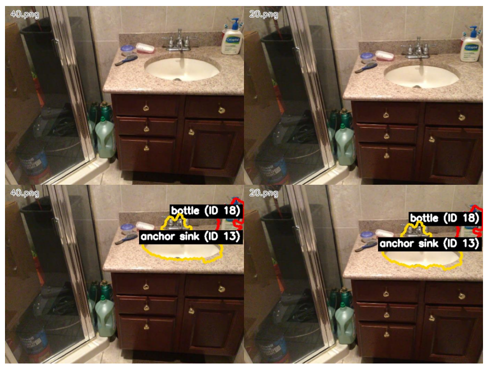
Symbolic Parser Output

```
{
  "json_obj": {
    "category": "bottle",
    "relations": [
      {
        "relation_name": "right",
        "objects": [
          {
            "category": "sink",
            "relations": []
          }
        ]
      }
    ]
  },
  "is_viewpoint_instruction": true,
  "viewpoint_anchor_phrases": ["sink"],
  "viewpoint_anchor_categories": ["sink
    "],
  "viewpoint_anchor_front_relation": "
    opposite_front"
}
```

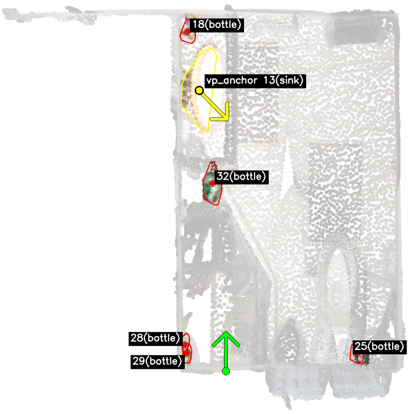
Executor Top-5 Selection

Rank	1	2	3	4	5
ID	25	32	28	29	18
Cat.	bottle	bottle	bottle	bottle	bottle
Score	1.000	0.905	0.905	0.905	3.54e-5

Visual Prompt



Instance-Centric Prompting. The candidate view centers on the correct target object.



Perspective-Aligned Prompting. Top-down map under the predicted observer viewpoint.

VLM Reasoning

```
{
  "reasoning": "The target description
    specifies 'the bottle on the right-
    hand side of the sink'. The top-down
     map shows the sink (ID 13) with
    bottle ID 18 positioned to its right.
     In the candidate image for ID 18,
    the bottle is clearly visible on the
     right side of the sink from the
    observer's viewpoint, consistent
    with the green arrow indicating the
    observer's direction. Other bottles
    (IDs 25, 28, 29, 32) are not located
     near the sink or on its right side.
     Thus, ID 18 best matches the
    description.",
  "object id": 18
}
```

C Baseline Configuration Summary

Table 8 summarizes the original configurations of the training-free methods in our main comparison. These methods differ in their object candidate sources, scene representations, and language and visual models; the main-table results should therefore be interpreted with these engineering differences in mind. Our matched-backbone comparisons are reported separately in Tables 1 and 5.

Method	Object candidate source	Candidate / scene representation	Language and visual models
OriGround	ScanNet-aligned instance objects (Nr3D); Mask3D proposals (ScanRefer)	Instance point clouds and boxes, orientation bases, multi-view crops, and a top-down map	Qwen-VL-Max; PointNeXt, Mask3D, and Orient-Anything-V2
LaSP (Mi et al., 2025)	ScanNet-aligned instance objects (Nr3D)	Instance 3D features and boxes, relation programs, and a candidate image grid	GPT-4o; PointNeXt
ZSVG3D (Yuan et al., 2024b)	ScanNet-aligned instance objects (Nr3D); Mask3D proposals (ScanRefer)	Instance 3D features and locations, paired 2D views, and executable visual programs	GPT-3.5; PointNeXt, Mask3D, and BLIP-2
SeeGround (Li et al., 2025)	ScanNet-aligned instance objects (Nr3D); Mask3D proposals (ScanRefer)	Object table and query-aligned scene renderings	Qwen2-VL-72B; Mask3D
VLM-Grounder (Xu et al., 2025)	Open-vocabulary 2D detections and masks projected into 3D	Stitched RGB views, grounded 2D masks, and multi-view projected 3D boxes	GPT-4o; Grounding DINO 1.5 and SAM-Huge
LLM-Grounder (Yang et al., 2024)	OpenScene / LERF open-vocabulary 3D candidates	Text-retrieved 3D feature candidates and spatial coordinates	GPT-4; OpenScene / LERF
OpenScene (Peng et al., 2023)	CLIP-aligned 3D feature field	Dense text-aligned point and voxel features	CLIP; OpenScene

Table 8: Original configurations of the training-free methods in the main comparison.

D Prompts

In this section, we show the key prompts used by OriGround. We organize them into two parts: symbolic prompts and reasoning prompts. Listings 1 and 2 present the two symbolic stages, namely relation decomposition and observer-viewpoint inference. Listing 3 presents the reasoning prompt for viewpoint-aligned visual grounding.

Prompt for semantic parsing.

```
You are a parser for 3D referring expressions.

## Objective
Convert a natural-language query into a normalized relation tree with a target
    object category and a set of canonical spatial predicates.

## Output Schema
Return STRICT JSON only:
{
  "category": "target category",
  "relations": [...]
}

Each relation entry must use:
- "relation_name": canonical predicate
- "objects": referenced object list
- "negative": optional boolean

## Predicate Inventory
Use only the following relation names.
```

```
Unary:
["corner", "on the floor", "against wall", "smaller", "larger", "taller", "lower
    ", "within"]

Binary:
["above", "below", "beside", "close", "far", "left", "right", "front", "behind",
     "across"]

Ternary:
["between", "center", "middle"]

## Parsing Rules
1. The root "category" is the target object category named by the query.
2. Normalize surface wording to the closest canonical predicate above.
3. Never invent a new relation name.
4. Keep predicates simple and reusable. Do not create relation strings such as "
    left of a blue box" or "facing the window".
5. Encode nested descriptions recursively inside the referenced objects.
6. For unary relations, use "objects": [].
7. For binary relations, provide exactly one referenced object.
8. For ternary relations, provide exactly two referenced objects.
9. If the sentence explicitly negates a relation, set "negative": true for that
    relation.
10. Ignore observer-viewpoint phrases here; they are handled in the next
    symbolic stage.

## Recommended Procedure
- Identify the target category.
- Extract the minimal set of spatial relations needed to distinguish the target.
- Build referenced objects recursively.
- Reason internally if needed, but output JSON only.

## Example
Sentence: The correct whiteboard is the one on a table.

Output:
```json
{
  "category": "whiteboard",
  "relations": [
    {
      "relation_name": "above",
      "objects": [
        {
          "category": "table",
          "relations": []
        }
      ]
    }
  ]
}
```

3 more examples...
```

Prompt for observer-viewpoint inference.

```
You are doing stage 2 of NR3D parsing: observer-viewpoint inference only.

## Goal
Decide whether the sentence defines an explicit or implicit observer frame, and
    if so, extract the viewpoint anchor and its front-direction relation.

## Input
- sentence: {sentence}
- stage1_json_obj: {json_obj_text}
```

```
## Output
Return STRICT JSON only with the schema:
{
  "is_viewpoint_instruction": true,
  "viewpoint_extraction_reason": "short reason",
  "viewpoint_anchor_phrases": ["phrase1"],
  "viewpoint_anchor_categories": ["category1"],
  "viewpoint_anchor_front_relation": "same_front"
}

## Core Constraints
1. Use the sentence as the primary evidence.
2. Use `stage1_json_obj` only as frozen context for the target category and relation anchors. Do not rewrite the stage-1 symbolic parse.
3. Preserve the stage-1 target category. Do not switch the target to an anchor category.
4. Keep `viewpoint_extraction_reason` short.
5. Anchor phrases should be copied from the sentence when possible.
6. Anchor categories should be normalized object categories usable downstream.

## Positive Cases
Set `is_viewpoint_instruction=true` for explicit observer-frame expressions such as:
- `facing ...`
- `looking at ...`
- `looking toward ...`
- `head on`
- `from the foot of the bed`
- `if you are sitting on ...`

Also mark `true` for implicit group-ordering cases when the sentence resolves one object in a group using an ordering or directional frame, for example:
- `the middle box of the stack`
- `the top one`
- `the blanket above the other blanket`
- `the narrower of the two stalls`
- `the back pillow on the right`
- `the last on the right`

When no explicit external anchor exists in such group-ordering cases, it is acceptable to use the target category itself as `viewpoint_anchor_categories`.

## Front-Relation Policy
Valid values of `viewpoint_anchor_front_relation` are only: `same_front`, `opposite_front`, `left_of_front`, `right_of_front`, `unknown`

Typical `same_front` cases:
- `sitting on ...`
- `sitting in ...`
- `through the doorway`
- some `standing at ...` cases where the observer shares the anchor's front direction

Typical `opposite_front` cases:
- `facing ...`
- `looking at ...`
- `head on`
- many group-ordering cases without an explicit external anchor

## Negative Cases
Pure object-object relations usually stay `false` when they do not define an observer frame or a group-ordering frame, for example:
- `farthest from the door`
- `next to the desk`
- `close to the window`

When `is_viewpoint_instruction=false`, output:
```

```
- `viewpoint_anchor_phrases: []`
- `viewpoint_anchor_categories: []`
- `viewpoint_anchor_front_relation: "unknown"`

## Examples
Example 1
sentence: Facing the desk, the book on the left.
stage1_json_obj:
{"category":"book","relations":[{"relation_name":"left","objects":[{"category":"
    desk","relations":[]}]}]}
output:
{
  "is_viewpoint_instruction": true,
  "viewpoint_extraction_reason": "Facing the desk defines a fixed observer
    direction.",
  "viewpoint_anchor_phrases": ["the desk"],
  "viewpoint_anchor_categories": ["desk"],
  "viewpoint_anchor_front_relation": "opposite_front"
}

3 more examples...
```

Prompt for viewpoint-aligned visual grounding.

```
You are a visual grounding expert for indoor scenes.

## Input
You will receive multiple images.

## Image Order
- The first image is a TOP-DOWN annotation-only map: plain background with vector
     overlays(availability: {has_topdown_map}, type: {topdown_map_kind}).
- The remaining images are one per candidate object.

## Sample Type
For this sample type, the instruction contains a fixed observer viewpoint cue.

## Top-Down Map Semantics
Top-down is annotation-only (without_pcd). The semantics are:
- GREEN arrow(s) encode hypothesized observer viewing direction.
- YELLOW arrow near reference anchor indicates anchor front direction estimate.
- RED / YELLOW regions indicate candidate / anchor boundaries.
- MAGENTA regions indicate overlap objects that are both candidate and anchor.
- Candidates are labeled as `(ID x) category`; anchors are labeled as `anchor (
    ID x) category`.
- Candidate/anchor centers are marked with red/yellow circles for relative left/
    right/front/behind comparison.

## Candidate Image Layout
Each candidate image has a 2x2 layout:
- Top row: two original full-frame views (raw scene context)
- Bottom row: the same two views with annotations

## Candidate Annotation Semantics
Annotation semantics (critical):
- The candidate is highlighted with a red translucent region + red boundary
    contour and labeled as `category (ID x)`.
- Anchor/reference objects (if visible in the same frame) are highlighted and
    labeled as `anchor <category> (ID x)`.
- Anchor annotations are relational cues only, not automatic targets.
- Do NOT confuse candidate and anchor: red annotation indicates candidate; yellow
     annotation indicates anchor-only reference.

## Task
Select exactly one object ID from the candidate list that best matches the
    target description.

## Input Fields
```

```
Target Object Description: "{utterance}"
Candidate list: {candidates}

## Decision Policy
0) Candidate vs anchor disambiguation (mandatory)
   - First verify annotation color semantics before reasoning: candidate is red;
     anchor is non-red reference annotation.
   - Never output an object only because it appears as an anchor marker.

1) Scene-image-first grounding
   - Candidate images (raw + annotated views) are primary evidence.
   - Top-down is auxiliary for global horizontal layout and viewpoint-
    conditioned directional cross-check.

2) Fixed-viewpoint alignment from language (critical)
   - Parse the sentence for the viewpoint cue (e.g., facing/looking at/standing
    at the foot of the bed).
   - Use top-down directional cues to instantiate the observer frame, then
    evaluate left/right/front/back.
   - If directional cues conflict, prefer the interpretation most consistent
    with the sentence viewpoint cue and candidate-view evidence.

3) Top-down reliability handling
   - Orientation arrows can be noisy; allow small geometric tolerance.
   - If the top-down directional signal strongly conflicts with candidate-view
    evidence, trust candidate-view evidence.

4) Top-down limitation
   - Top-down supports horizontal floor-plane reasoning only.
   - Do not use top-down for vertical relations such as on/under, above/below, or
     height.

5) Final selection
   - Choose exactly one candidate ID with the strongest joint evidence from
    category, relation, and multi-view consistency.
   - Always output one ID from Candidate list, even under uncertainty.

## Output Format
- Return STRICT JSON only (no markdown, no extra text):
{
  "reasoning": "brief evidence-focused explanation",
  "object id": 0,
  "confidence": 0.0
}
```

For queries without viewpoint instructions, we use the same global-context reasoning template but do not instantiate a fixed observer frame; the top-down map is then treated only as auxiliary global context for candidate distribution and anchor relations, rather than as a committed directional frame.

E AI Assistance Statement

Generative AI assistants were used for language polishing, selected aspects of project code implementation, and discussions of design alternatives. The authors led the research process and reviewed and validated all AI-assisted outputs. All methodological decisions, experiments, result interpretation, scientific claims, and final manuscript content remain the responsibility of the authors.